SCIENTIFIC JOURNAL PUBLICATIONS (as First Author) (19)

2022 **Sheikh, M. U.,** Lempiainen, J, and Jantti, R., ***"Arguments for One Radio Access Network (OneRAN) Mobile Infrastructure",*** in Springer Journal on Telecommunication Systems, Issue 80, Page 477-486, 2022. https://doi.org/10.1007/s11235-022-00917-y

2021 **Sheikh, M. U.,** Xie, B., Ruttik, K., Yigitler, H., Jantti, R., and Hamalainen, J., ***"Ultra Low Power Wide Range Backscatter Communication Using Cellular Generated Carrier",*** in MDPI Sensor Journal, Special issue Wireless Sensor Networks and IoT for Smart Cities, 2021. DOI: https://doi.org/10.3390/s21082663

2021 **Sheikh, M. U.,** Ruttik, K., Jantti, R., and Hamalainen, J., ***"A Study about Signal Variation with Minor Receiver Displacement in a Meeting Room at 60 GHz: Measurements and Simulations",*** in EURASIP Journal on Wireless Communications and Networking. (Status: Accepted)

2021 **Sheikh, M. U.,** Badihi, B., Ruttik, K., and Jantti, R., ***"Bluetooth 5.0 Performance Analysis with Adaptive Physical Layer: Measurements and Simulations",*** in Hindawi Journal on Wireless Communications and Mobile Computing. (Status: Accepted)

2019 **Sheikh, M. U.,** Biswas, R, Lempiainen, J, and Jantti, R., ***"Assessment of Coordinated Multipoint Transmission Modes for Indoor and Outdoor Users at 28 GHz in Urban Macrocellular Environment",*** in Advances in Science, Technology and Engineering System Journal. Volume 4, Issue 2, Page 119-126, 2019. DOI: 10.25046/aj040216

2018 **Sheikh, M. U.,** Hiltunen, K., and Lempiainen, J. ***"Enhanced Outdoor to Indoor Propagation Models and Impact of Different Ray Tracing Approaches at Higher Frequencies",*** in Advances in Science, Technology and Engineering System Journal. Volume 3, Issue 2, Page 58-68, 2018. DOI: 10.25046/aj030207

2018 **Sheikh, M. U.,** and Lempiainen, J. ***"Analysis of Outdoor and Indoor Propagation at 15 GHz and Millimeter Wave Frequencies in Microcellular Environment",*** in Advances in Science, Technology and Engineering System Journal. Volume 3, Issue 1, Page 160-167, 2018. DOI: 10.25046/aj030120.

2017 **Sheikh, M. U.,** Sae, J., and Lempiainen, J. ***"Arguments of Innovative Antenna Design and Centralized Macro Sites for 5G",*** in Springer Journal on Wireless Personal Communications, Volume 96, Issue (2017), Page 6007-6019. DOI: 10.1007/s11277-017-4460-4.

2017 **Sheikh, M. U.,** and Lempiainen, J. ***"Analysis of Multipoint Transmission at Higher Frequencies using 3D Ray Tracing",*** in Wiley International Journal of Communication Systems, Volume 30, Issue 13, September 2017, doi.org/10.1002/dac.3274.

2016 **Sheikh, M. U.,** and Lempiainen, J., ***"Impact of Antenna Radiation Pattern Variation on the Performance of SPMA at 28 GHz",*** in

Springer Journal on Wireless Networks. DOI: 10.1007/s11276-016-1338-4

2016 **Sheikh, M. U.,** and Lempiainen, J., ***"Macro Cellular Network Transition from Traditional Frequency Range to 28 GHz Millimeter Wave Frequency Band",*** in Springer Journal on Wireless Networks. DOI: 10.1007/s11276-016-1390-0

2016 **Sheikh, M. U.,** and Lempiainen, J., ***"Impact of Non-Typical Optimum Antenna Configurations on Traditional and Higher Order Sectored Sites in Dense and Ultra Dense Networks",*** in IGI International Journal of Interdisciplinary Telecommunications and Networking (IJITN), Volume 9, Issue 3, pp. 21-44.

2016 **Sheikh, M. U.,** Sae, J., and Lempiainen, J., ***"In Preparation Towards 5G Networks: The Impact of Macro Site Densification and Sector Densification on System Capacity",*** in Springer Journal on Telecommunication Systems, Volume 65, Issue 4, pp. 621-636, August 2017.

2015 **Sheikh, M. U.,** Sae, J., and Lempiainen, J., ***"Evaluation of SPMA and Higher Order Sectorization for Homogeneous SIR through Macro Sites",*** in Springer Journal on Wireless Networks, Volume 22, Issue 4, pp. 1079-1091, May 2016. DOI:10.1007/s11276-015-1019-8.

2014 **Sheikh, M. U.,** and Lempiainen, J., ***"Will New Antenna Materials enable Single Path Multiple Access (SPMA)?",*** in Springer Journal on Wireless Personal Communications, Volume 78, Issue 2(2014) September, Page 979-994. DOI: 10.1007/s11277-014-1796-x.

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SCIENTIFIC CONFERENCE PUBLICATIONS (as First Author) (39)

2023 **Sheikh, M. U.**, Ali, M., Carpintero, G., Ruttik, K., Mutafangwa, E., and Jantti, R., ***"Channel Characterization at Sub-THz Band with Measurements and Ray Tracing in Indoor Case"*** IEEE Consumer Communications and Networking Conference (CCNC), Las Vegas, USA, 8-11 January, 2023.

2022 **Sheikh, M. U.**, Ruttik, K., Padilla, E. Y., Mutafungwa, E., Jantti, R., and Hamalainen, J., ***"X-Haul Solutions for Different Functional Split Options Using THz and Sub-THz bands"*** in ACM 25th International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MWSim'22), Montreal, Canada, 24-28 October, 2022. (Status: Under review**).**

2022 **Sheikh, M. U.**, Ali, M., Carpintero, G., Ruttik, K., Mutafangwa, E., and Jantti, R., ***"Power Angular Measurements and Ray Tracing Simulations at Sub-THz Frequencies in Corridor"*** IEEE Wireless Communications and Networking Conference (WCNC), 2022.

2022 **Sheikh, M. U.,** Lempiainen, J, and Jantti, R., ***"Capacity Limitation of Small Cell Densification"*** in Proc. 36th International Conference on Information Networking (ICOIN'22), Bangkok, Thailand, 12-15 January, 2022.

2021 **Sheikh, M. U.**, Mela, L., Norshahida, S., Ruttik, K., and Jantti, R., ***"Outdoor to Indoor Path Loss Measurement at 3.5 GHz, 6.5 GHz, and 26.5 GHz Commercial Frequency Bands"*** in 24th IEEE International Symposium on Wireless Personal Multimedia Communications, Okayama, Japan, 13-16 December, 2021.

2021 **Sheikh, M. U.**, Norshahida, S., Mela, L., Salo, J., Ruttik, K., and Jantti, R., ***"Measurement Based Study of Large Scale Propagation Path Loss and Coverage for Commercial 5G Frequencies in Urban Macro Cellular Environment"*** in 17th IEEE International Conference on Wireless and Mobile Computing, Networking and Communications (WiMOB), Virtual, 11-13 Oct, 2021.

2021 **Sheikh, M. U.**, Jantti, R., and Hamalainen, J., ***"Distributed Antenna System in 3GPP Specified Industrial Environment"*** in IEEE Vehicular Technology Conference, VTC Spring, 2021, Helsinki, 25-28 April, 2021.

2021 **Sheikh, M. U.**, Ruttik, K., Saba, N., Mutafungwa, E., Jantti, R., and Hamalainen, J., ***"Analysis of Drone Propagation with Ray Tracing from sub-6 GHz Upto Terahertz Frequencies in a Real World Urban Environment"*** in Proc. IEEE International Conference on Information Networking (ICOIN'21), Jeju Island, Korea, 13-16 January.

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2020 **Sheikh, M. U.**, Riaz, M., Jameel, F., Sharma, N., Sharma, V., Alazab, M., and Jantti, R., ***"Quality-Aware Trajectory Planning of Cellular Connected UAVs"*** in 2nd Workshop on Drone Assisted Wireless Communications for 5G and Beyond co-located with ACM 26th Annual International Conference on Mobile Computing and Networking, London, 21-25 September.

2020 **Sheikh, M. U.**, Ruttik, K., Jantti, R., and Hamalainen, J ***"Measurements and Ray Tracing Simulations: Impact of Different Antenna Positions on Meeting Room Coverage at 60 GHz"*** in Proc. IEEE 29th European Conference on Networks and Communications (EuCNC), Dubrovnik, Croatia, June, 2020.

2020 **Sheikh, M. U.**, Hajj, M. E , Jantti, R., Hamalainen, J., Zaharia, G., Zein, G., Sadek, S., and Farhat., H ***"Channel Characterization and Validation of Ray Tracing Simulations with Measurements at 60 GHz in Office Environment"*** in Proc. IEEE 27th International Conference on Telecommunications (ICT), Bali, Indonesia, April, 2020.

2020 **Sheikh, M. U.**, Jantti, R., and Hamalainen, J., ***"Performance Comparison of Ray Tracing and 3GPP Street Canyon Model in Microcellular Environment"*** in Proc. IEEE 27th International Conference on Telecommunications (ICT), Bali, Indonesia, April, 2020.

2020 **Sheikh, M. U.**, Jameel, F., Yigitler, H., Wang, X., and Jantti, R., ***"Monostatic Backscatter Communication in Urban Microcellular Environment Using Cellular Networks"*** IEEE Wireless Communications and Networking Conference (WCNC), 2020.

2020 **Sheikh, M. U.**, Jantti, R., and Hamalainen, J., ***"Impact of Interference Suppression under Ray Tracing and 3GPP Street Canyon Model"*** IEEE Vehicular Technology Conference, VTC Spring, 2020. (Status: Accepted**).**

2020 **Sheikh, M. U.,** Asghar, M. Z., and Jantti, R., *"**Dual Connectivity in Non-Stand Alone Deployment mode of 5G in Manhattan Environment",*** in IEEE International Conference on Electronics, Information, and Communication (ICEIC'20). **(Bronze Best Paper Award)**

2019 **Sheikh, M. U.**, Hamalainen, J., Gonzalez, D., Jantti, R., and Gonsa, O., ***"Usability Benefits and Challenges in mmWave V2V Communications: A Case Study",*** IEEE International Conference on Wireless and Mobile Computing, Networking and Communications (WiMOB), 2019.

2019 **Sheikh, M. U.**, Ruttik, K. and Jantti, R., *"**DAS and UDN Solutions for Indoor Coverage at Millimeter Wave (mmWave) Frequencies",*** IEEE Vehicular Technology Conference, VTC Fall, 2019.

2019 Sheikh, M. U., **Ghavimi, F.,** Ruttik, K. and Jantti, R., *"**Drone Detection and Classification Using Cellular Network: A Machine Learning Approach",*** IEEE Vehicular Technology Conference, VTC Fall, 2019.

2019 **Sheikh, M. U**., Ruttik, K. and Jantti, R., ***"Performance Analysis of Vertical and Higher Order Sectorization in Urban Environment at 28 GHz",*** in Proc. IEEE 26th International Conference on Telecommunications (ICT), Hanoi, Vietnam, April, 2019.

2019 **Sheikh, M. U**., Ghavimi, F., Ruttik, K. and Jantti, R., ***"Analysis of Indoor Solutions for Provision of Indoor Coverage at 3.5 GHz and 28 GHz for 5G System",*** in Proc. IEEE 26th International Conference on Telecommunications (ICT), Hanoi, Vietnam, April, 2019. **(Second Best Paper Award)**

2019 **Sheikh, M. U**., Ruttik, K. and Jantti, R., ***"Performance Evaluation of Switched Beam Antenna with Different Configurations at 28 GHz",*** IEEE Wireless Communications and Networking Conference (WCNC), 2019.

2019 **Sheikh, M. U**., Duan, R. and Jantti, R., ***"Validation of Backscatter Link budget Simulations with Measurements at 915 MHz and 2.4 GHz",*** IEEE Vehicular Technology Conference, VTC Spring 2019.

2018 **Sheikh, M. U**., Sae, J., and Lempiainen, J. ***"Multipath Propagation Analysis of 5G Systems at Higher Frequencies in Courtyard (Small Cell) Environment",*** IEEE 1st 5G World Forum 2018.

2018 **Sheikh, M. U.,** Biswas, R., and Lempiainen, J. ***"Performance Evaluation of Coordinated Multipoint Transmission at 28 GHz Frequency using 3D Ray Tracing",*** in IEEE Vehicular Technology Conference (VTC) Spring 2018.

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2017 **Sheikh, M. U.,** Hiltunen, K., and Lempiainen, J. ***"Angular Wall Loss Model and Extended Building Penetration Model for Outdoor to Indoor Propagation",*** in Proc. 13th IEEE International Wireless Communication and Mobile Computing (IWCMC), pp. 1291-1296, June 2017, Valencia, Spain.

2016 **Sheikh, M. U.,** and Lempiainen, J., ***"Migration to 28 GHz Frequency with Higher order Sectorization in Urban Macro Cellular Environment",*** in Proc. IEEE 23rd International Conference on Telecommunications (ICT), pp. 1-5, 2016, Thessaloniki, Greece. doi: 10.1109/ICT.2016.7500460

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2011 **Sheikh, M. U.,** and Lempiainen, J., ***"Impact of Smart Traffic Handling Scheme on Load Balancing and Interference Margin n Multilayer Network with UMTS900",*** IEEE International Wireless Communication and Mobile Computing (IWCMC'11). Paper Status (Accepted) to be published in July 2011.

2011 **Sheikh, M. U.,** Jagusz, R., and Lempiainen, J., ***"Performance Evaluation of Adaptive MIMO Switching in Long Term Evolution",*** IEEE International Wireless Communication and Mobile Computing (IWCMC'11). Paper Status (Accepted) to be published in July 2011

2011 **Sheikh, M. U.,** and Lempiainen, J., *"**Impact of Smart Traffic Handling Scheme on Downlink Power of UMTS Base station in Multilayer Network**",* In Proc. IEEE International Conference of Ultra Modern Telecommunications, (ICUMT'11), Budapest, Hungary, October 2011.

2011 **Sheikh, M. U.,** Jagusz, R., and Lempiainen, J., *"**LTE performance with variable number of users per Transmission Time Interval along with AMS**",* In Proc. IEEE International Conference of Ultra Modern Telecommunications, (ICUMT'11), Budapest, Hungary, October 2011.

2009 **Sheikh, M. U.,** and Lempiainen, J., ***"Impact of Penalty Time on Multilayer Network along with UMTS900 Deployed with Smart Traffic Handling Scheme",*** In Proc. IEEE Student Conference on Research and Development (SCOReD 2009), Malaysia, November 2009.

2009 **Sheikh, M. U.,** and Lempiainen, J.,***"UMTS900 deployment with different call handling strategies"*,** In Proc. IEEE International Symposium on Wireless Communication Systems (ISWCS'09), Siena-Tuscany, Italy, September 2009.

Book Chapter (as First Author) (2)

2019 **Sheikh, M. U.,** Lempiainen, J., and Jantti, R., ***"Multidimensional Analysis of LTE Network Roll out with Typical and Non-Typical Antenna Configuration",*** in book on Strategic Innovations and Interdisciplinary Perspectives in Telecommunications and Networking by IGI global.

2017 **Sheikh, M. U.,** Sharareh, N., and Lempiainen, J., ***"SPMA: A Centralized Macrocellular Approach for Enhancing System Capacity of Future Smart Networks",*** in book on Information Innovation Technology in Smart Cities. (DOI:10.1007/978-981-10-1741-4_15)

SCIENTIFIC JOURNAL PUBLICATIONS (as Co-Author) (1)

2020 Duan, R., Wang, X., Yigitler, H., **Sheikh, M. U.,** Jantti, R., and Han, Z., ***"Ambient Backscatter Communications for Future Ultra-low-power Machine Type Communications: Solutions, Challenges, Opportunities, and Future Research Trends",*** in IEEE Communications Magazine. (Status: Accepted).

CONFERENCE PUBLICATIONS (as Co-Author) (10)

2022 Norshahida, S., Mela, L., **Sheikh, M. U.**, Salo, J., Ruttik, K., and Jantti, R., ***"Millimeter-Wave Radio Link Analysis for 5G FWA by Combining Measurements and Geospatial Data"*** in IEEE International Conference on Communications, Seoul, South Korea, 16-22 May, 2022. (Status: Under review).

2022 Al-Nahari, A., Mela, L., Jantti, R., and **Sheikh, M. U.**, ***"Artificial Rich Scattering-Assisted MIMO System Using Passive Backscatter Devices"*** in 3rd Information Communication Technologies Conference (ICTC), Nanjing, China, 6-8 May, 2022. (Status: Accepted).

2021 Biswas, R., **Sheikh, M. U.**, Yigitler, H., Lempiainen, J., and Jantti, R., ***"Interference Analysis of Bi-static Backscatter Communication System: Two Backscatter Devices"*** in IEEE International Conference on RFID Technology and Applications (RFID-TA), Delhi, India, 6-8 October, 2021. (Status: Accepted**).**

2021 Norshahida, S., Mela, L., **Sheikh, M. U.**, Ruttik, K., Salo, J., and Jantti, R., ***"Rural Macrocell Path Loss Measurements for 5G Fixed Wireless Access at 26 GHz"*** in IEEE 5G World Forum 2021, Montreal, Canada, 13-15 October, 2021. (Status: Accepted**).**

2021 Biswas, R., **Sheikh, M. U.**, Yigitler, H., Lempiainen, J., and Jantti, R., ***"Interference Suppression Requirements for Bistatic Backscatter Communication System"*** in IEEE Vehicular Technology Conference, VTC Spring, 2021, Helsinki, 25-28 April, 2021. (Status: Accepted**).**

2021 Norshahida, S., Mela, L., **Sheikh, M. U.**, Ruttik, K., and Jantti, R., ***"Measurements at 5G Commercial 26 GHz Frequency with Above and Rooftop Level Antenna Masts in Urban Environment "*** in IEEE Vehicular Technology Conference, VTC Spring, 2021, Helsinki, 25-28 April, 2021. (Status: Accepted**).**

2020 Jameel, F., **Sheikh, M. U**., and Jantti, R., ***"Enabling Finite Blocklength Industrial IoT Networks: Industry 4.0 under Blockages",*** IEEE Vehicular Technology Conference, VTC Fall 2020. (Status: Accepted)

2020 Badihi, B., **Sheikh, M. U.**, Ruttik, K., and Jantti, R., ***"On Performance Evaluation of BLE 5 in Indoor Environment: A Experimental Study",*** IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), 2020. (Status: Accepted**).**

2019 Badihi, B., Liljemark, A., **Sheikh, M. U.**, Lietzen, J, and Jantti, R., ***"Link Budget Validation for Backscatter-Radio System in Sub-1GHz",*** IEEE Wireless Communications and Networking Conference (WCNC), 2019**.**

2019 Ahmed, Y., and **Sheikh, M. U.,** ***"Linear Array Processing using Modified Singular Value Decomposition",*** IEEE International Symposium on Recent Advances in Electrical Engineering (RAEE), 2019, Islamabad, Pakistan.

Book Chapter (as Co-Author) (1)

2015 Yunas, S. F., Sae, J., and **Sheikh, M. U.,** ***"Planning Guidelines and Principles for 5G RAN*****",** in book on 5G Networks: A Research and Development Perspective.